

YANG SUN

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EDUCATION

University of Arizona
Ph.D. in Astronomy and Astrophysics

Tucson, USA
2021 - Present

Beijing Normal University
B.S in Astronomy

Beijing, China
2016 - 2020

POSITIONS

Graduate Researching Assistant

2021 - Present

Graduate Teaching Assistant

Fall 2023, Fall 2025

RESEARCH INTERESTS

Extragalactic Astronomy, Black Hole-Galaxy Coevolution, AGN and SF Feedback, Galaxy Quenching

PUBLICATIONS

See the full publication list on ADS

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Total Publications: 31, with citations of 431 (Sept. 2025)

First Author Publications: 7, with citations of 83

First Author Papers:

1. **Sun, Y.**, Ji, Z., Rieke, G. H., et al. (2025), Extreme Neutral Outflow in an Inactive Quenching Galaxy at $z \sim 1.3$, arXiv e-prints, arXiv:2504.14682.
2. **Sun, Y.**, Rieke, G. H., Lyu, J., et al. (2025), Evolution of the M_*M_{BH} Relation from $z \sim 6$ to the Present Epoch, *The Astrophysical Journal*, 983, 165.
3. **Sun, Y.**, Lyu, J., Rieke, G. H., et al. (2025), No Evidence for a Significant Evolution of $M_{BH} - M_*$ Relation in Massive Galaxies up to $z \sim 4$, *The Astrophysical Journal*, 978, 98.
4. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al. (2024), Evolution of gas flows along the starburst to post-starburst to quiescent galaxy sequence, *Monthly Notices of the Royal Astronomical Society*, 528, 5783.
5. **Sun, Y.**, Lee, G.-H., Zabludoff, A. I., et al. (2023), Evolution of Gas Flows over the Starburst to Post-Starburst to Quiescent Galaxy Sequence, Resolving the Rise and Fall of Star Formation in Galaxies, 373, 193.
6. **Sun, Y.**, Yuan, H., & Chen, B. (2022), Validations and Corrections of the SFD and Planck Reddening Maps Based on LAMOST and Gaia Data, *The Astrophysical Journal Supplement Series*, 260, 17.
7. **Sun, Y.**, Deng, D.-S., & Yuan, H.-B. (2021), Precision of the Chinese Space Station Telescope (CSST) stellar radial velocities, *Research in Astronomy and Astrophysics*, 21, 092.

Significant Contribution Papers:

1. Zhu, Y., Bonaventura, N., **Sun, Y.**, et al. (2025), SMILES Data Release II: Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpect Medium-Resolution Spectra, arXiv e-prints, arXiv:2508.12599.

2. Deng, D., **Sun, Y.**, Wang, T., Wang, Y., & Jiang, B. (2022), Infrared Excess of a Large OB Star Sample, *The Astrophysical Journal*, 935, 175.
3. Yuan, H.-B., Deng, D.-S., & **Sun, Y.** (2021), A star-based method for precise wavelength calibration of the Chinese Space Station Telescope (CSST) slitless spectroscopic survey, *Research in Astronomy and Astrophysics*, 21, 074.
4. Deng, D., **Sun, Y.**, Jian, M., Jiang, B., & Yuan, H. (2020), Intrinsic Color Indices of Early-type Dwarf Stars, *The Astronomical Journal*, 159, 208.

Co-author Papers:

1. D'Eugenio, F., Helton, J. M., Hainline, K., **including Sun, Y.**, et al. (2025), JADES and SAPPHIRES: galaxy metamorphosis amidst a huge, luminous emission-line region, *Monthly Notices of the Royal Astronomical Society*, 542, 960.
2. Rinaldi, P., Rieke, G. H., Wu, Z., **including Sun, Y.**, et al. (2025), Beyond the Dot: an LRD-like nucleus at the Heart of an IR-Bright Galaxy and its implications for high-redshift LRDs, arXiv e-prints, arXiv:2507.17738.
3. Stone, M. A., Rieke, G. H., Lyu, J., **including Sun, Y.**, et al. (2025), The $z = 7.08$ quasar ULAS J1120+0641 May Never Reach a "Normal" Black Hole to Stellar Mass Ratio, arXiv e-prints, arXiv:2507.13489.
4. Rieke, G. H., Buiten, V. A., Goldberg, C. E., **including Sun, Y.**, et al. (2025), Low Accretion Rates in Black Holes in Late-stage Merger Ultraluminous Infrared Galaxies, *The Astrophysical Journal*, 988, 17.
5. Zhu, Y., Rieke, M. J., Ji, Z., **including Sun, Y.**, et al. (2025), A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images, *The Astrophysical Journal*, 986, 162.
6. Zhu, Y., Alberts, S., Lyu, J., **including Sun, Y.**, et al. (2025), SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions, *The Astrophysical Journal*, 986, 18.
7. Lin, X., Fan, X., Sun, F., **including Sun, Y.**, et al. (2025), The Large-scale Environments of Low-luminosity AGNs at $3.9 < z < 6$ and Implications for Their Host Dark Matter Halos from a Complete NIRCам Grism Redshift Survey, arXiv e-prints, arXiv:2505.02896.
8. Zhang, J., Egami, E., Sun, F., **including Sun, Y.**, et al. (2025), Abundant Population of Broad H Emitters in the GOODS-N Field Revealed by CONGRESS, FRESCO, and JADES, arXiv e-prints, arXiv:2505.02895.
9. Lin, X., Egami, E., Sun, F., **including Sun, Y.**, et al. (2025), The Luminosity Function and Clustering of H Emitting Galaxies at $z \approx 4 - 6$ from a Complete NIRCам Grism Redshift Survey, arXiv e-prints, arXiv:2504.08028.
10. Zhu, Y., Rieke, M. J., Ho, L. C., **including Sun, Y.**, et al. (2025), Nuclear Winds Drive Cold Gas Outflows on Kiloparsec Scales in Reionization-Era Quasars, arXiv e-prints, arXiv:2504.02305.
11. D'Eugenio, F., Hainline, K., Arribas, S., **including Sun, Y.**, et al. (2025), Forever Blowing Bubbles: What Powers a 24-kpc Ionized Gas Nebula Around a Normal Galaxy at $z=6$?, JWST Proposal. Cycle 4, 7335.
12. Scholtz, J., Parlanti, E., Carniani, S., **including Sun, Y.**, et al. (2025), Tentative rotation in a galaxy at $z \sim 14$ with ALMA, arXiv e-prints, arXiv:2503.10751.
13. Hainline, K. N., Maiolino, R., Juodbalis, I., **including Sun, Y.**, et al. (2025), An Investigation into the Selection and Colors of Little Red Dots and Active Galactic Nuclei, *The Astrophysical Journal*, 979, 138.
14. Alberts, S., Lyu, J., Shivaiei, I., **including Sun, Y.**, et al. (2024), SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging, *The Astrophysical Journal*, 976, 224.
15. Rinaldi, P., Bonaventura, N., Rieke, G. H., **including Sun, Y.**, et al. (2024), Not Just a Dot: the complex UV morphology and underlying properties of Little Red Dots, arXiv e-prints, arXiv:2411.14383.
16. Zhu, Y., Bakx, T. J. L. C., Ikeda, R., **including Sun, Y.**, et al. (2024), Discovery of a Unique Close QuasarDSFG Pair Linked by a [C II] Bridge at $z = 5.63$, *Research Notes of the American Astronomical*

Society, 8, 284.

17. Alberts, S., Williams, C. C., Helton, J. M., **including Sun, Y.**, et al. (2024), To High Redshift and Low Mass: Exploring the Emergence of Quenched Galaxies and Their Environments at $3 < z < 6$ in the Ultra-deep JADES MIRI F770W Parallel, The Astrophysical Journal, 975, 85.

18. Shivaiei, I., Alberts, S., Florian, M., **including Sun, Y.**, et al. (2024), A new census of dust and polycyclic aromatic hydrocarbons at $z = 0.72$ with JWST MIRI, Astronomy and Astrophysics, 690, A89.

19. Ji, Z., Williams, C. C., Rieke, G. H., **including Sun, Y.**, et al. (2024), Extended hot dust emission around the earliest massive quiescent galaxy, arXiv e-prints, arXiv:2409.17233.

20. Wevers, T., Coughlin, E. R., Pasham, D. R., **including Sun, Y.**, et al. (2023), Live to Die Another Day: The Rebrightening of AT 2018fyk as a Repeating Partial Tidal Disruption Event, The Astrophysical Journal, 942, L33.

OBSERVING PROPOSALS

ALMA Cycle 12	2025.1.01045.S (7.2h)	PI
ALMA Cycle 12	2025.1.01067.S (45.6h)	co-I
JWST Cycle 4	GO 7335 (14.2h)	co-I

TALKS

NOIRLab Flash Talk, Tucson, USA	<i>Sept. 2025</i>
CRISOL2025 (flash talk), Toledo, Spain	<i>May 2025</i>
JWST/NIRCam Science Team Meeting, Tucson, USA	<i>March 2025</i>
Dust and Gas throughout Cosmic Time, Hiroshima, Japan	<i>Dec. 2024</i>
JWST/MIRI Meeting, Tucson, USA	<i>Oct. 2024</i>
STSCI Spring Symposium (flash talk), Baltimore, USA	<i>Apr. 2024</i>
Steward Internal Symposium, Tucson, USA	<i>Sept. 2023</i>
Columbia University Seminar, New York, USA	<i>July 2023</i>
Galaxy Formation Workshop, Tucson, USA	<i>Apr. 2023</i>
IAU Symposium 373, Busan, South Korea	<i>Aug. 2022</i>

SERVICE AND OUTREACH

Astronomy Tutoring Offered for Majors and Minors (ATOMM) program in Steward Observatory	<i>2025</i>
Volunteer for Black Holes Camp in Flandrau Planetarium	<i>2025</i>
Co-organizer of Steward Internal Symposium	<i>2025</i>
Co-organizer of Perspective Graduate Student Visit in Steward Observatory	<i>2024</i>
Member of International Scholars Task Force in Steward Observatory	<i>2021-Present</i>

SKILLS

Programming: Python, Fortran, MATLAB

Software: CASA, TOPCAT, LATEX, EXCEL

Language: English, Chiense